Dead neurons resurrect by an entry–exit law: the smoothed-ReLU limit of training dynamics at an activation boundary

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Abstract

For shallow softplus networks under gradient flow we show that, in the sharp limit $\beta \to \infty$ of the activation stiffness, the death and resurrection of a neuron is governed by a hybrid system *with memory*: the dead neuron's parameters freeze, and reactivation occurs when the signed time integral of the network residual returns to zero — the classical entry–exit function of slow–fast dynamics — with rate $O(\beta^{-1})$. The switching face carries an identically zero field on the dead side, a degeneracy outside generic piecewise-smooth exit theory and precisely of the exponentially flat type of Kristiansen; the coupling of the forcing to the flow itself (the residual is generated by the very trajectory it gates) has, to our knowledge, no analogue in the entry–exit literature. The law is independent of the regularization within monotone sigmoid classes, including a GELU-type erf smoothing. All constants are verified numerically; companion pre-registered experiments probe the stochastic gate under SGLD.

1 Setup and hypotheses

Data $(x_i, y_i)_{i=1}^n$ in general position, shallow network $f_\theta(x) = \sum_k g_\beta(w_k x + b_k)$ with $g_\beta(z) = \beta^{-1}\log(1 + e^{\beta z})$ (softplus; see §5 for general sigmoids), quadratic loss $L = \frac{1}{2n}\sum_i (f_\theta(x_i) - y_i)^2$, gradient flow $\dot\theta = -\nabla L$. Fix a neuron A and a datum j; write $z = \theta_A \cdot (x_j, 1)$ for its margin, $c = \|(x_j, 1)\|^2/n$, and $R(t) = y_j - f_{-A}(x_j; \theta_B(t))$ for the residual *excluding* A. The *hybrid reference* θ^{hyb} runs the full ReLU flow until the first time t_{hit} with $z = 0$, then freezes θ_A and runs θ_B alone; set R^{hyb}, $A^{hyb}(t) = \int_{t_{hit}}^t R^{hyb}$.

Hypotheses. **(H1)** $R^{hyb}(t_{hit}) < 0$ (transversal death; this also gives the transversal arrival, Lemma 3). **(H2)** A^{hyb} has a first zero at $t_{exit} < T$ with $R^{hyb}(t_{exit}) > 0$ and $A^{hyb} < 0$ on (t_{hit}, t_{exit}). **(H3)** the other margins of A stay $\le -\delta_1 < 0$ on the episode. **(H4)** no other switching event occurs in $[0, T]$: the margins of the remaining neurons stay bounded away from zero along the hybrid reference, the activation pattern of A is constant on $[0, t_{hit})$, and no second episode of A begins before T.

2 Main result

Definition 1 (hybrid limit)**.** *θ^∞ equals: the ReLU flow on $[0, t_{hit}]$; on $[t_{hit}, t_{exit}]$, $\theta_A \equiv \theta_A(t_{hit})$ (frozen) and θ_B the B-only flow; on $[t_{exit}, T]$, the ReLU flow restarted at $(\theta_A(t_{hit}), \theta_B^{hyb}(t_{exit}))$.*

Theorem 1 (T2′-min)**.** *Under (H1)–(H4), for every $\delta > 0$,*

$$\sup_{t\in[0,T]\setminus(B_\delta(t_{hit})\cup B_\delta(t_{exit}))} \|\theta^\beta(t) - \theta^\infty(t)\| = O_\delta\Big(\tfrac{\log\beta}{\beta}\Big), \qquad t^\beta_{release} = t_{exit} + O(\beta^{-1}),$$

where $t^\beta_{release}$ is the first return of the margin z to 0 after death. The $\log\beta$ is sharp and is carried solely by the normal coordinate of θ_A inside the episode (the sticking depth of Lemma 4); θ_B, the tangential components of θ_A, and $t^\beta_{release}$ obey the cleaner rate $O(\beta^{-1})$. The limit is that of the regularized flow: discrete ReLU gradient descent freezes the dead neuron forever (the limits do not commute).

3 Proof outline

Six lemmas; each has an executable verification script in the repository.

Lemma 1 (normal form)**.** *In the gauge $w = e^{\beta z}$, fast time $\tau = \beta t$: $dw/d\tau = cR(t)\,w^2/(1+w)\,(1+E_\beta)$, $|E_\beta| \le \log(1+w)/(\beta|R|)$ — the quadratic normal form of De Maesschalck–Schecter with $\varepsilon = \beta^{-1}$. With $u = 1/w$ and $\Phi(u) = \log u + u$ one has the exact quadrature $d\Phi/dt = -c\beta R(t)(1+E_\beta)$.*

Lemma 2 (B-subsystem and feedback)**.** *On any episode interval where $w \le w_{\max}$, the coupling of A into the B-field is $\le \beta^{-1}\log(1+w_{\max})$ and $\sup\|\theta^\beta_B - \theta^{hyb}_B\| \le C\beta^{-1}$ (Grönwall). In the bulk of sticking $u = O(\beta)$, so the coupling there is $O(\beta^{-2})$.*

Lemma 3 (entry corner)**.** *$\dot z^0(t^-_{hit}) = cR^{hyb}(t_{hit}) < 0$ by (H1); by comparison in the window $z \in [\beta^{-1}\log w_0, \delta_z]$, the crossing of $\{w = w_0\}$ occurs at $t_{hit} + O(\beta^{-1})$ with state $O(\beta^{-1})$-close to $\theta^\infty(t_{hit})$.*

Lemma 4 (sticking and release, prescribed R)**.** *For prescribed R satisfying (H1)–(H2), with $A(t) = \int_{t_{hit}}^t R$: the exact additive identity $d\Phi/dt = -c\beta R + c\log(1+w)$ holds (no division by R; the interior zero of R forced by (H1)–(H2) via the intermediate value theorem is harmless — the layer is held by the* area*, not the sign of R), and on compact subintervals of (t_{hit}, t_{exit}), $\Phi(u(t)) = -c\beta A(t)(1+o(1))$, hence $u(t) = c\beta|A(t)|(1+o(1))$ (depth $z \sim -\beta^{-1}\log\beta$), and u returns to $O(1)$ at $t_{exit} + O(\beta^{-1})$ — no $\log\beta$ factor. Sub/super solutions from the monotonicity of the flow in R give explicit constants; perturbations δ of R shift t_{exit} by $\le c\,\delta\,(t_{exit} - t_{hit})/|A'(t_{exit})| + O(\delta^2)$.*

Lemma 5 (coupled bootstrap)**.** *The set of times up to which "θ_B is $2M\beta^{-1}$-close to the hybrid and the margin stays in the layer" is open and closed in $[t_{cross}, t_{exit} - \epsilon]$: under the assumption, coupling is $O(\beta^{-1})$ (corners) / $O(\beta^{-2})$ (bulk), so Lemma 2 returns the better constant $M\beta^{-1}$ and Lemma 4 with $\delta = O(\beta^{-1})$ keeps the layer. Hence the closed-loop law $t^\beta_{release} = t_{exit} + O(\beta^{-1})$.*

Lemma 6 (exit corner; scalar reduction)**.** *Post-release the field is transversal outward ($cR^{hyb}(t_{exit}) > 0$, (H2)); gluing as in Lemma 3. Under (H3) the other data contribute $O(e^{-\beta\delta_1})$ and the face-j field is rank one along $(x_j, 1)$, so the tangential components of θ_A freeze to exponential accuracy and the episode reduces to the scalar problem.*

4 Numerical verification

Instance $X = (0.5, 1.5)$, $Y = (0.5, 1.0)$, $\theta_A(0) = (-0.5, 0.27)$, $\theta_B(0) = (1.144, 0.261)$; $t_{hit} = 0.1114$, $t_{exit} = 1.1325$, $\delta_1 = 0.508$. Selected checks (scripts `passo0`–`passo6`): entry $t_{cross} = 0.11141 + 7.13/\beta$ (fit residual 0.2% over $\beta \in [3 \cdot 10^2, 10^4]$); depth ratio $u_{\max}/(c\beta|A|_{\max})$ increasing $0.888 \to 0.982$ toward the predicted limit 1 (prescribed-R variant of the instance, whose own $t_{exit} = 1.1352$); bulk feedback $\times\beta^2 \approx 43$ constant over 1.5 decades; closed-loop $\beta\,(t^{\beta}_{release} - t_{exit}) \to -14.4$ (a definite first-order coefficient, sharper than the theorem); tangential drift of θ_A at machine zero.

5 Regularization independence; discussion

The release criterion depends only on $\int R\,dt = 0$, not on the sigmoid: for the erf class ($g'_\beta = \Phi_{\text{gauss}}(\beta z)$, a GELU-type smoothing; GELU itself, $z\Phi_{\text{gauss}}(z)$, has non-monotone derivative and is not a member of the family for any β) the coupled system releases at the same t_{exit} with $O(\beta^{-1})$ gap (verified). Open: the cone apex (corank ≥ 2), where weight decay produces genuine sliding toward the most singular stratum; the SGLD gate (pre-registered experiments show the deterministic schedule is falsified at first order by persistent residual fluctuations — the stochastic entry–exit is the object of the next cycle); and the non-commutation of the two limit orders: the *field-first* selection (flow of the pointwise-limit generalized gradient) freezes degenerate neurons forever — this is the rigorous branch of Eberle, Jentzen, Riekert and Weiss [8], whose selected solution has non-decreasing degenerate set — whereas our *flow-first* limit ($\beta \to \infty$ of trajectories) resurrects by the area law. The two orders do not commute; cf. [8] for the field-first side (their degenerate stratum is the fully-zero one, not literally our functionally dead neuron — the convention $\text{relu}'(0) = 0$ and the no-revival mechanism are the same). Conceptual anchors: entry–exit function [1, 11]; flat slow manifolds [2]; generic exit and regularized-limit theory [4, 3]; nonsmooth training formalism [5]; ReLU–LCP equivalence [6]; dying ReLU [7]; the canonical regularization of discontinuous fields is Sotomayor–Teixeira [9, 10] — whose generic regions require both one-sided normal components nonzero, so the dead face (one side identically zero) is outside its hypotheses by construction.

Provenance

Every lemma except the exit-corner half of Lemma 6 has a dedicated executable symbolic/numeric verification committed alongside (the exit corner is verified indirectly through the release times). Experiments were adversarially reviewed (two rounds) and pre-registered by commit before execution; the referee of this draft questioned the audit trail and the check was re-run: the pre-registration commit precedes the confirmatory run in the repository history. Two earlier experimental designs and one earlier conjecture were rejected by the same process and are archived. All numerics are a single two-neuron instance; breadth is future work.

References

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